

Numerically Controlled Oscillator on PIC[®] Microcontrollers

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INTRODUCTION

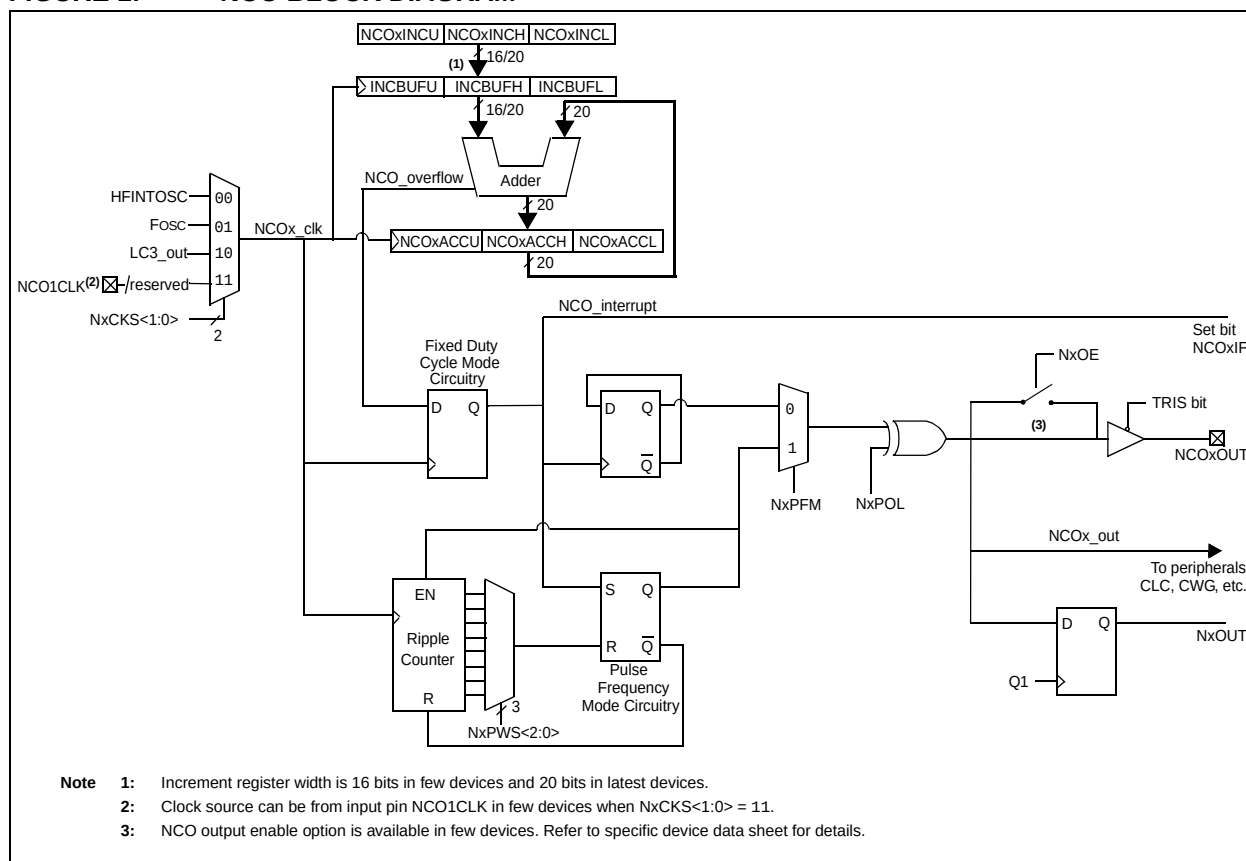
Applications that require precise linear frequency control such as lighting control, tone generators, radio tuning circuitry, fluorescent ballast and class D audio amps can be designed using Microchip's Numerically Controlled Oscillator (NCO) peripheral. The NCO offers a simplification in design for such applications by providing a programmable precision linear frequency generator ranging from less than 1 Hz to 1 MHz on devices with a 16-bit increment register, or less than 1Hz to 32 MHz on devices with a 20-bit increment register. This document briefly describes the NCO and its configuration.

OVERVIEW OF THE NUMERICALLY CONTROLLED OSCILLATOR (NCO)

The NCO uses the Direct Digital Synthesis (DDS) technique to generate waveforms. DDS is a technique used for generating an analog waveform, generally of a sinusoidal wave shape, from a time varying signal in its digital form and a DAC.

The NCO module operates on the principle of DDS by repeatedly adding a fixed value to an accumulator (Figure 1). The addition is done with carry and the remainder is left in the accumulator after the overflow. Accumulator overflow is the raw NCO output. The NCO behaves as a 20-bit timer that uses the overflow of the accumulator to divide the input frequency and create an output signal.

FIGURE 1: NCO BLOCK DIAGRAM



The raw NCO output can be further modified by stretching the pulse using a delay circuit as shown in [Figure 1](#) (i.e., Pulse Frequency mode circuitry) or by toggling a flip-flop. The modified NCO output is then connected internally to other peripherals and optionally output to an I/O pin. The NCO output can trigger an internal interrupt when the accumulator overflows. The NCO period changes in discrete steps to create an average frequency.

The overflow of the NCO module is based on the formula listed in [Equation 1](#).

EQUATION 1: NCO ACCUMULATOR OVERFLOW FREQUENCY

$$F_{\text{OVERFLOW}} = \left(\frac{\text{NCO Clock Frequency} \times \text{Increment Value}}{2^n} \right)$$

Note: n = 20 = Accumulator width in bits

With such an arrangement, the NCO output response will be very linear across a wide range of frequencies, ranging from 0 kHz up to 1 MHz using an oscillator clock of 16 MHz and 16-bit increment register and from 0 kHz to 32 MHz using 32 MHz oscillator clock and 20-bit increment register. The frequency resolution that can be obtained is precise. For example, if an input clock of 1 MHz is used, then a frequency resolution of 0.95 Hz is obtained. If the input clock is 100 kHz, then the frequency increments become 0.095 Hz. The linear frequency control and the increased frequency resolution are the key distinguishing factors when compared to the traditional PWM-based frequency control.

NCO Clock

Clock sources available to the NCO are HFINTOSC, Fosc, LCxOUT, and in some devices, the NCO1CLK pin. The NCO clock source is selected by configuring the NxCKS<2:0> bits in the NCOxCLK register. The HFINTOSC selection will continue to run even if the device is put into Sleep mode and will continue to consume power.

Accumulator

The accumulator is a 20-bit register with a maximum value of 1,048,575. When the accumulator overflows, the output of the NCO module will change state. Read and write access to the accumulator is available through three registers:

- NCOxACCL
- NCOxACCH
- NCOxACCU

Adder

The NCO adder is a full adder, which operates independently from the system clock. It adds the value of the increment value to the accumulator on each NCO clock pulse. The result is placed back in the accumulator.

Increment Value

The increment value can be up to 20 bits. Some devices support increment values of up to 16 bits only. The Increment registers are double-buffered to allow for value changes to be made without first disabling the NCO module. The buffer loads are immediate when the module is disabled. Writing to the NCOxINCUI and NCOxINCH registers first is necessary because the buffer is loaded synchronously with the NCO operation after the write is executed on the NCOxINCL register.

NCO Output

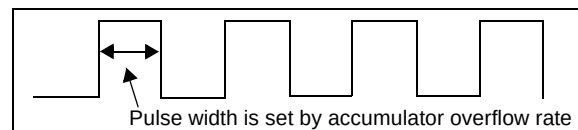
The output of the NCO has several options that can be set in the NCOCON register. The output can be enabled or disabled (NxOE bit) in some devices and also inverted (NxPOL bit) by setting or clearing bits in the NCOCON register. The output can also be monitored in software by reading the state of the NxOUT bit in the NCOCON register. The output can be connected internally to other peripherals such as CLC or CWG.

The NCO module can output a signal in one of two modes set by using the N1PFM bit in the NCOCON1 register:

- Fixed 50% Duty Cycle (FDC) with N1PFM = 0
- Pulse Frequency Modulation (PFM) with N1PFM = 1

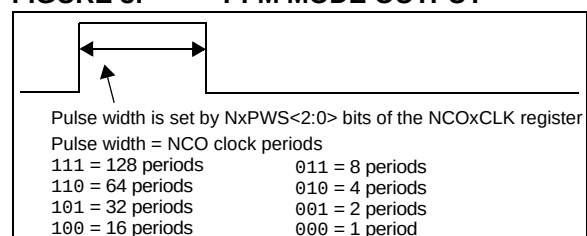
The Fixed Duty Cycle mode toggles the output on every accumulator overflow; as long as the adder value and clock do not change, this will result in a 50% duty cycle output (see [Figure 2](#)).

FIGURE 2: FDC MODE OUTPUT



The Pulse Frequency Modulation mode will trigger a pulse on every accumulator overflow. The active state of the output can vary in width by multiple clock periods set by three bits (NXPWS<2:0>) in the NCOCON register. Once the clock period expires, the output returns to an inactive state. This provides a pulsed output (see [Figure 3](#)).

FIGURE 3: PFM MODE OUTPUT

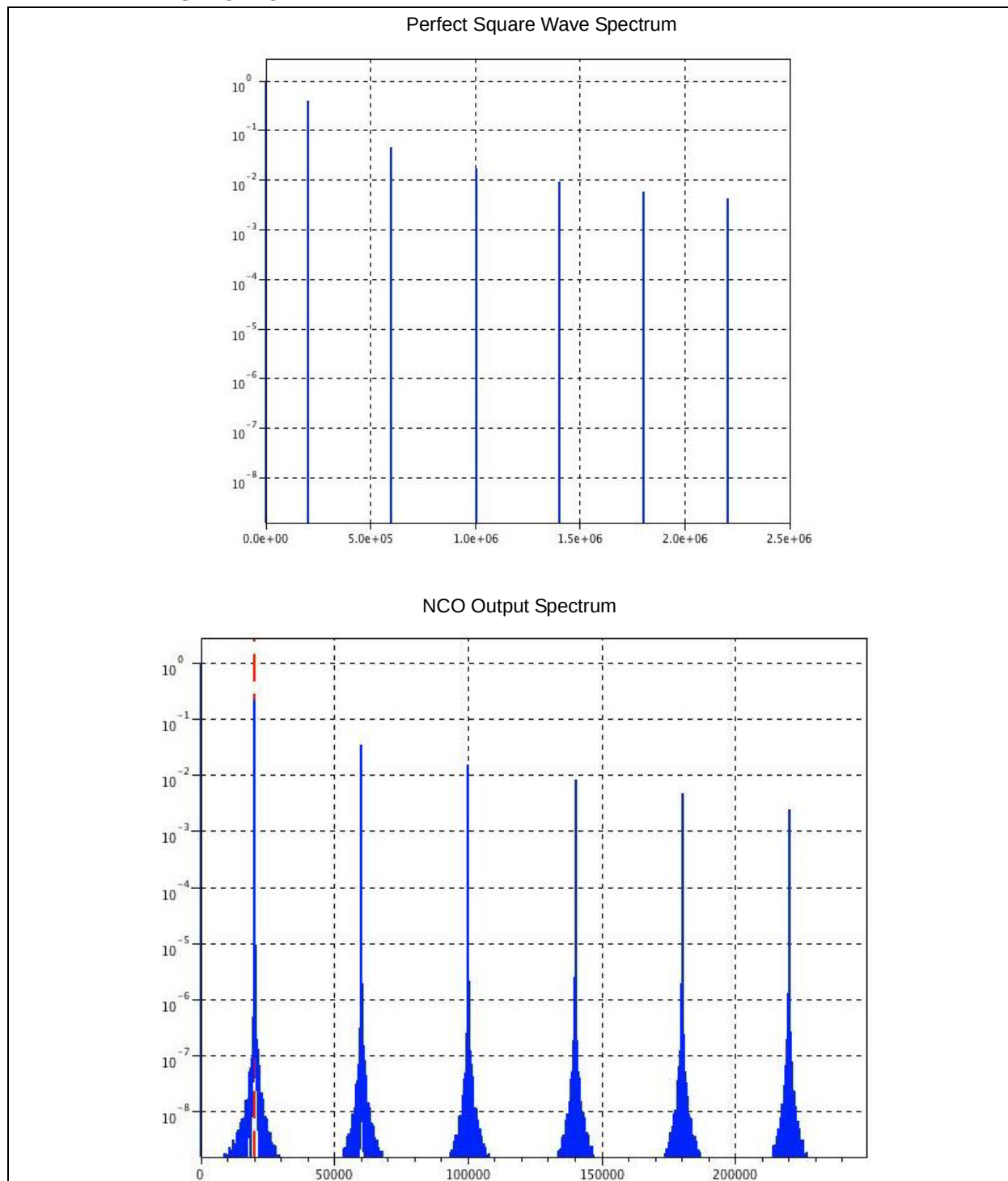


NCO AND DDS

The NCO module generates precisely controllable output frequencies using the DDS technique. The DDS technique essentially provides a clock with carefully controlled jitter on it. Therefore, it is necessary that the signal be aggregated on the frequency domain.

Figure 4 illustrates the typical output spectra when generating 50% duty cycle square wave using the NCO module. The sideband noise generated by the jitter is insignificant in comparison to the fundamental frequency. When plotted on a logarithmic scale, the NCO output compares to that of a perfect square wave.

FIGURE 4: PERFECT SQUARE WAVE SPECTRUM COMPARED TO THE NCO OUTPUT SPECTRUM



NCO OUTPUT AND PWM OUTPUT COMPARISON

When using a PWM module to generate a pulse train with variable frequency, the frequency is calculated as in [Equation 2](#).

EQUATION 2: PWM FREQUENCY CALCULATION

$$F_{PWM} = \frac{F_{OSC}}{4(PR2 + 1)}$$

Where,

F_{PWM} = Desired frequency of PWM

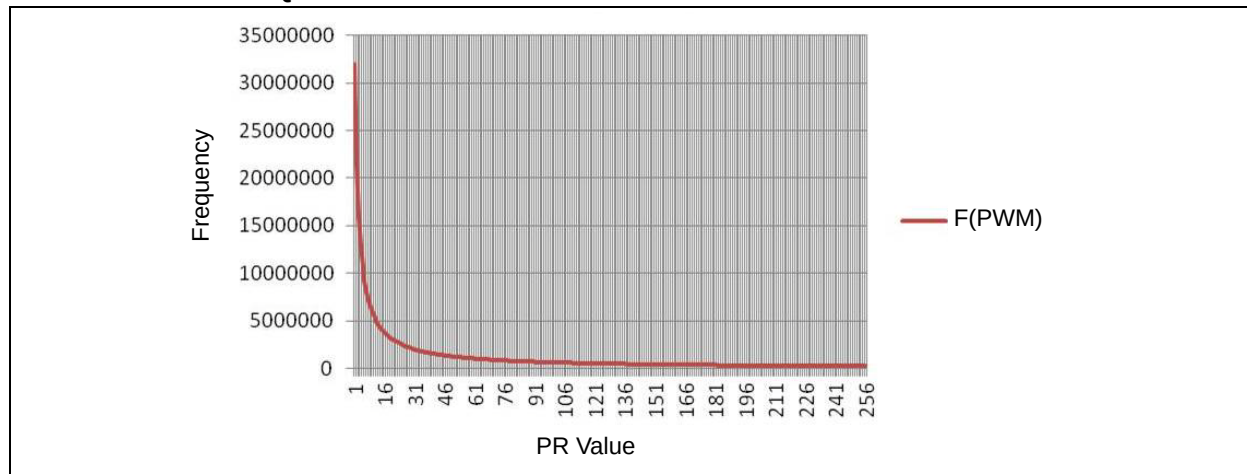
F_{OSC} = Oscillator clock frequency

$PR2$ = Period register to be loaded

For an 8-bit PR2 register, the value can vary from 0 to 255. With the oscillator clock frequency being fixed, the value in the PR2 register determines the frequency of the PWM output. Since the PR2 register value forms the denominator in [Equation 2](#), any change in the value of PR2 will not yield a linear variation of F_{PWM} , although the incremental change in the denominator or PR2 is linear.

[Figure 5](#) illustrates the variation of the PWM frequency with respect to a corresponding change in the PR2 value.

FIGURE 5: FREQUENCY vs. PR VALUE IN PWM MODULE



When NCO clock frequency is chosen as F_{OSC} , the relation between the frequency of the NCO output and the incremental register is provided in [Equation 3](#). From [Equation 3](#), note that F_{NCO} is directly proportional to the increment value, and the accumulator overflow value is always fixed to $2^{20} = 1048576$. Therefore, any change in the increment value will yield a very linear variation in the output frequency of the NCO (i.e., F_{NCO}).

EQUATION 3: OUTPUT FREQUENCY OF THE NCO

$$F_{NCO} = \left(\frac{F_{OSC}}{2^{20}} \right) (\text{Increment Value})$$

Where,

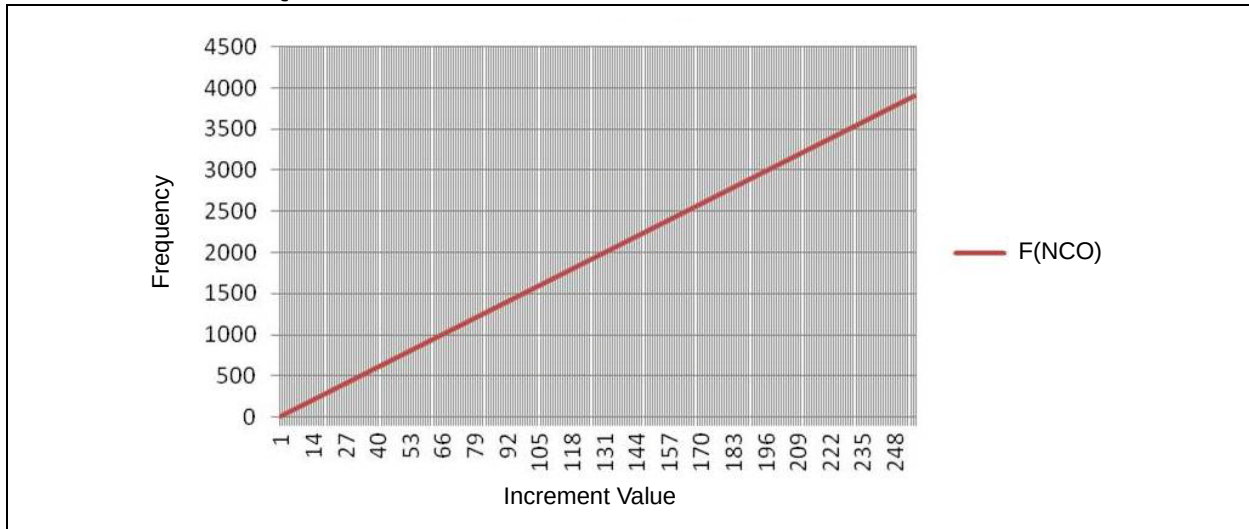
F_{NCO} = Frequency of the output of NCO module

F_{OSC} = Oscillator clock frequency

Increment Value = Value loaded to change F_{NCO}

This concept illustrates the variation of F_{NCO} with a corresponding change in the increment value (see Figure 6).

FIGURE 6: FREQUENCY vs. INCREMENT VALUE IN NCO MODULE



Therefore, a better frequency resolution over a wide frequency range can be obtained using the NCO for waveform generation, when compared to the conventional PWM-based approach.

CONFIGURATION OF THE NCO

Table 1 lists the registers associated with the NCO.

TABLE 1: REGISTERS ASSOCIATED WITH THE NCO

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
NCO1ACCU	—	—	—	—	NCO1ACC<19:16>			
NCO1ACCH	NCO1ACC<15:8>							
NCO1ACCL	NCO1ACC<7:0>							
NCO1CLK	N1PWS<2:0>			—	—	—	N1CKS<1:0>	
NCO1CON	N1EN	N1OE	N1OUT	N1POL	—	—	—	N1PFM
NCO1INC	—	—	—	—	NCO1INC<19:16>			
NCO1INCH	NCO1INC<15:8>							
NCO1INCL	NCO1INC<7:0>							

EXAMPLE 1: CODE SNIPPET FOR GENERATING 100 kHz FREQUENCY WITH 50% DUTY CYCLE USING NCO IN FIXED DUTY CYCLE MODE

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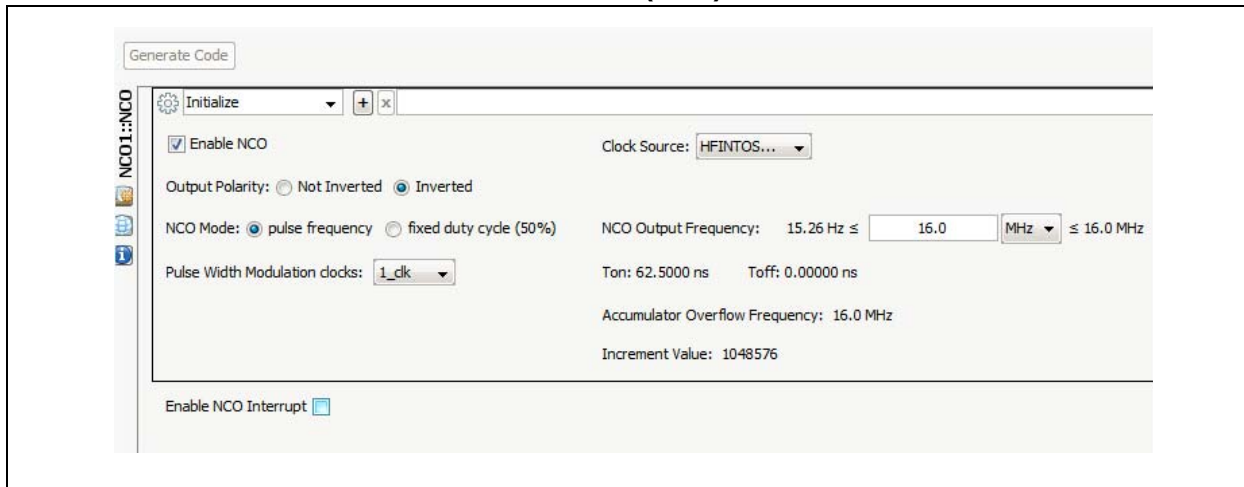
OSCCONbits.IRCF= 0b1111; //FOSC = 16 MHz
OSCCONbits.SCS = 0;      //Clock determined by FOSC<2:0> in Configuration Words
OSCCONbits.SPLEN = 0;    //4x PLL is disabled
NCO1CONbits.N1EN = 1;    //NCO module is enabled
NCO1CONbits.N1PFM = 0;    //NCO operates in Fixed Duty Cycle mode
NCO1CONbits.N1POL = 0;    //NCO output signal is active high (non-inverted)
NCO1CLKbits.N1CKS = 0b01; //NCO clock = FOSC = 16 MHz
NCO1INC = 0x3334;        //NCO increment value for
//NCO output toggled at frequency = 200 kHz to get the square wave of 100 kHz
//Only by changing the increment value any other frequency can be generated
//for example for 150 kHz frequency; NCO1INC = 0x4CCD;

```

USING MPLAB® CODE CONFIGURATOR (MCC) TO CONFIGURE NCO

The MPLAB® Code Configurator, which is a plug-in tool for MPLAB® X IDE, generates the drivers for controlling and driving the NCO modules based on the settings and selections made in the GUI.

FIGURE 7: MPLAB® CODE CONFIGURATOR (MCC) FOR NCO



NCO API Function List

The following APIs are generated for the NCO module by the MCC tool:

- `void NC01_Initialize (void)`: initializes the NCO1CON, NCO1CLK, NCO1ACC and NCO1INC registers and enables the NCO module
- `bool NC01_GetOutputStatus(void)`: returns the output status on the accumulator overflow
- `void NC01_ISR(void)`: Interrupt Service Routine where user can include any application specific code

CONCLUSION

This technical brief describes how the NCO peripheral functions on PIC microcontrollers. The NCO simplifies the design for applications requiring precise linear frequency control such as lighting control, tone generators, radio tuning circuitry, fluorescent ballast and class D audio amplifiers. The configuration of the NCO has been described in this technical brief along with the listing of APIs generated by the MCC tool. More details can be found in the product specific data sheet.

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